SpeakHER — Product Requirements Document

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Find the words to express your thoughts. Draft 1 for design and development handoff. Labels used throughout: **[Assumption]**, **[Recommendation]**, **[Validate]**, **[Future]**.

1. Executive summary

SpeakHER is a private, scenario-based practice space where girls and young women (16–25) entering tech rehearse explaining their ideas by voice or text, get specific AI feedback, revise, and try again. The MVP does one thing well: **choose a scenario → respond by voice or text → receive specific AI feedback → try again → review progress.**

Decision	Recommendation
Core bet	Repeatable, low-stakes rehearsal with specific feedback helps users structure and explain ideas. This is a hypothesis to validate, not a fact.
MVP size	6 scenarios, voice and text, one feedback screen, revise-and-retry, simple history. No streaks, leaderboards, community or live human coaching.
AI posture	Observations over scores. Two strengths, one or two improvements, one retry prompt. No overall "confidence score".
Privacy posture	Private by default. Raw audio is not stored after transcription. Transcripts are kept only until the user deletes them or after a short default retention period.
Minors	Users aged 16–17 are in scope. Age gate, plain-language consent, no public profiles, and legal review before launch.

Decision	Recommendation
Brand	Playful digital notebook meets creative studio: Lilac, Butter Yellow, Coral, Warm Cream, Deep Plum; Fredoka plus DM Sans; speech-bubble and hand-drawn-mark graphic language.
Build path	Clickable Figma prototype first with scripted feedback, then a functional MVP with a thin backend that proxies the AI and speech-to-text providers.

SpeakHER does not claim to fix confidence, remove structural barriers in tech, or guarantee interviews or jobs. It offers accessible, repeatable practice and useful feedback, and it sits alongside mentors, teachers and human feedback rather than replacing them.

Biggest uncertainties (detailed in section 18): whether target users want to practise this way, whether AI feedback is trusted and accurate enough, whether speech-to-text handles Nigerian-English and other accents fairly, and what the privacy and minors requirements are in the markets you launch in.

2. Product vision and positioning

Element	Statement
Vision	A world where any young woman exploring tech has a private place to rehearse her ideas until the words feel like hers.
Mission	Give girls and young women accessible, repeatable practice and specific feedback for explaining their thinking in learning, building and early-career situations.
Value proposition	Practise the conversations that come with learning and entering tech, in private, as often as you like, and get feedback you can act on in minutes. No performance, no audience, no judgement.

Element	Statement
Positioning statement	For girls and young women learning, building and applying in tech, SpeakHER is a communication practice tool that turns real situations (explaining a project, asking a question, answering an interview prompt) into short rehearsals with specific feedback. Unlike generic chatbots or interview-question lists, it is scenario-structured, voice-or-text, built around revision, and designed with privacy and minors in mind.

Primary segment [Assumption]: 18–25-year-olds in university, bootcamps, tech communities or early job searches who regularly need to explain projects or answer interview questions.

Secondary segment [Assumption]: 16–17-year-olds in coding clubs, school tech programmes and community cohorts (e.g. girls-in-tech programmes), likely reached through an organisation. Also: community leaders and facilitators who recommend practice tools to cohorts (a distribution channel, not a build target).

User needs: a safe place to be imperfect; concrete (not vague) feedback; relevant scenarios; the option not to speak aloud; control over their data; permission to sound like themselves.

Main use cases: project explanations, self-introductions, asking questions in class or events, explaining concepts to non-technical people, internship interview prep, networking intros, team-discussion contributions, describing a challenge or achievement.

Differentiators [Recommendation]:

- Scenarios drawn from tech learning and entry points, not generic public speaking.
- Revision loop is the centre of the product, not a one-shot score.
- Feedback that quotes the user's own words and refuses to invent details.
- Explicit fairness rules for accents, dialects and multilingual speech.
- Private-by-default design and plain-language data controls, including for minors.
- A distinct brand: a playful notebook-studio, not a corporate interview simulator.

Out of scope: fixing structural barriers (hiring bias, unequal access, hostile environments); technical skill teaching; replacing mentors, teachers or career coaches; guaranteeing confidence, interviews or employment; clinical anxiety support; accent

"correction"; social or community features in the MVP.

Positioning guardrail: SpeakHER should never imply that women are absent from tech because of their own communication. Copy frames practice as a resource, not a remedy for a deficit.

3. Problem statement and assumptions

Working problem statement [Assumption]: Many girls and young women entering tech have few low-stakes opportunities to practise putting technical or personal ideas into words before they must do so in classes, communities, interviews or teams. Practice with a knowledgeable, patient person is unevenly available, and self-practice usually lacks feedback.

What is not claimed: This PRD cites no statistics, survey results or user quotes, because none have been collected. Participation gaps in tech have many causes, including discrimination, unequal access to opportunity, hostile learning environments and lack of mentorship. Communication practice addresses only one slice.

Situations where the problem may occur [Assumption]

- Being asked "tell me about your project" and rambling or under-explaining.
- Hesitating to ask a question in a class, workshop or community event.
- Explaining a technical concept to a non-technical person.
- Internship or entry-level interviews, especially for first-timers.
- Introducing oneself at a networking event.
- Contributing an idea in a team discussion.

Existing alternatives and workarounds [Validate]

Alternative	Strength	Likely gap
Practising in the mirror or alone	Free, private	No feedback
Friends, peers, family	Warm, human	Feedback may be vague, biased, or unavailable; embarrassment
Mentors, teachers, career offices	High-quality, human	Scarce, scheduled, not equally accessible

Alternative	Strength	Likely gap
Generic AI chatbots	Available instantly	No structure, no scenarios, feedback varies, privacy unclear
Interview-question lists and videos	Cheap, plentiful	Passive; no response, no feedback
Public-speaking clubs, workshops	Community, human	Higher stakes, schedule, cost, location

Why scenario-based practice might help [Assumption]

Specific situations lower the blank-page problem; repeating the same scenario makes progress visible; separating "trying" from "being judged" may reduce pressure. All three need validation with real users.

What SpeakHER can and cannot do

Can realistically improve	Cannot solve
Structuring an explanation; adding specifics; matching the audience; familiarity through repetition	Structural bias, unequal access to opportunity, hostile environments
Giving users a private starting point for preparation	Real-time social anxiety, live conversation dynamics, body language
Making practice available outside office hours	Replacing human feedback, mentoring or sponsorship

Assumptions to validate before a full build

#	Assumption	How to test
A1	The target group has real, recurring situations where explaining ideas is difficult	Interviews
A2	They would use a tool rather than practise with people, and would return	Interviews, prototype test, 2-week diary

#	Assumption	How to test
A3	Voice is wanted for at least some practice	Prototype test with both modes
A4	AI feedback is perceived as specific, fair and useful	Wizard-of-Oz or prototype with real responses
A5	Speech-to-text is accurate enough for Nigerian-English and other accents to give fair feedback	Test set with diverse speakers
A6	Parents, guardians and organisations accept use by 16–17-year-olds	Interviews with facilitators; legal review
A7	Users will accept AI processing of their practice with the right disclosures	Concept test of the consent flow

Lightweight research plan (5–8 target users)

Recruit: 6–8 people aged 16–25: at least 2 aged 16–17 (with guardian or organisation consent), a mix of students, bootcamp learners, community members and early job seekers, with varied accents and first languages. 45 minutes each, remote.

Part 1, past experiences (no product mentioned):

1. Tell me about the last time you had to explain something you built or learned to someone else. What happened?
2. Walk me through the last time you wanted to ask a question in a class or event. What did you do?
3. When you have an upcoming conversation that matters (interview, presentation), how do you prepare? Show me if you can.
4. What did you try last time you wanted to get better at this? What helped? What didn't?
5. Who do you ask for feedback, and what makes you hesitant to ask?
6. Tell me about a time something went well. What made it easier?
7. When have you avoided a speaking situation? What was the cost?
8. What do you use now on your phone or computer to prepare, if anything?

Part 2, after past behaviour is covered: show a low-fidelity prototype (3 scenarios, scripted feedback). Ask: "What do you expect this to do?", "What would you do next here?", "What would make you not trust this feedback?", "What would you not want recorded?"

Avoid: "Would you use this?", "Don't you find speaking hard?", hypotheticals about future use. Ask about the last time something happened.

How findings could change scope

If we hear…	Then…
Voice is rarely wanted	Lead with text; make voice a P1
Interview prep is the main pain	Narrow the MVP to interview and self-intro scenarios
Users mostly practise via community or cohort	Add facilitator or cohort features at P1; reposition as an organisation tool
AI feedback is distrusted	Add "show evidence" quotes, compare-with-example, mentor-shareable summary
Accent anxiety is the main issue	Make fairness and transcript review prominent; consider text-first onboarding
16–17s cannot use it without guardians	Launch 18+ and pursue partner-led access for under-18s

4. Target users and personas

[Assumption] All three personas are fictional hypotheses for design and must be revised after interviews. They contain no invented statistics or quotes.

	Amara, 17	Tolu, 21	Wanjiru, 24
Context	Secondary school; joined a coding club; building a small app	Final-year university student; building projects; applying for internships	Career-switcher finishing a bootcamp; entry-level job search
Situations	Presenting her project to the club; asking questions at a workshop	Project walkthroughs; internship interviews; networking events	Interviews; describing past work experience in tech terms

	Amara, 17	Tolu, 21	Wanjiru, 24
Needs	Low-pressure start; guardian-friendly privacy; text option	Structure for explaining projects; repeated practice before interviews	Translating past experience into relevant stories; concise answers
Concerns	"Who can see this?"; being judged for how she speaks	Sounding "too casual" or "not corporate"; generic feedback	Time; feedback that is vague or contradicts advice she has had
Design implications	Age-appropriate onboarding; text-first path; no public elements	Revise-and-retry loop; scenario variety	Short sessions; clear before/after comparison

Not personas (but must be considered): facilitators of girls-in-tech programmes who recommend the tool; guardians of 16–17-year-olds; users with speech, hearing, vision or motor disabilities; multilingual users who think in one language and present in another.

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5. Jobs to be done and user stories

#	Job to be done (when… I want to… so I can…)
J1	When I have to explain my project soon, I want to rehearse it and see where it loses people, so I can give a clearer explanation.
J2	When I want to ask a question or contribute, I want to try wording it first, so I can speak up with less hesitation.
J3	When I am preparing for an interview, I want realistic practice questions and honest feedback, so I can improve my answers before they count.
J4	When I explain tech to a non-technical person, I want to check I am not using jargon, so I can be understood.

#	Job to be done (when… I want to… so I can…)
J5	When I am not ready to be heard by anyone, I want to practise in private, so I can experiment without being judged.
J6	When I have practised before, I want to see my earlier attempts next to the new one, so I can tell whether I am improving.
J7	When I am uncomfortable with recording my voice, I want to type instead, so I can still practise.
J8	When I am done, I want to delete my practice and my account, so I stay in control of my data.

User stories (by theme)

ID	As a…	I want…	So that…	Priority
US1	New user	to pick goals during onboarding	scenarios suit what I am preparing for	P0
US2	User	to choose a scenario and read who I am talking to and what is expected	I know what a good answer attempts	P0
US3	User	to answer by voice or by typing	I practise how I choose	P0
US4	Voice user	to review and correct my transcript before feedback	transcription errors do not distort feedback	P0
US5	User	feedback on what worked and one or two things to change	I know what to do next	P0
US6	User	to revise my answer and compare attempts	I can see what changed	P0
US7	User	to say a piece of feedback is wrong or unhelpful	the system does not mislead me	P0

ID	As a…	I want…	So that…	Priority
US8	Returning user	to see my history and repeat a scenario	I can track and practise again	P0
US9	User	to control recordings, transcripts and my account	my data is mine	P0
US10	User	to set a time goal or reminder	I build a habit on my terms	P1
US11	User	to practise in other languages or switch feedback language	I can practise in the language I will use	P2
US12	Facilitator	to share a scenario set with a cohort	my group practises the same material	P2

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6. Goals, non-goals, and success metrics

Product goals

1. Users complete a first practice session in under 10 minutes from sign-up.
2. Users find the feedback specific and fair enough to act on.
3. Users who start a scenario attempt it again with a revised answer.
4. Users keep control of their data and can delete it without friction.

Non-goals (MVP)

- Proving that SpeakHER improves communication ability (it will collect early signals only).
- Community, social sharing, leaderboards, streaks, badges or public profiles.
- Pronunciation, accent or fluency scoring.
- Live conversation or video avatars.
- Human coach marketplace.
- Native mobile apps.

Metric	Definition	Type	Early target [Assumption]
Onboarding completion	% of sign-ups reaching the dashboard	Usage	75%+
First practice completion	% of new users who submit a first response within 7 days	Usage	55%+
Retry rate	% of sessions with at least one revised attempt	Usage and engagement	40%+
Return practice	% of users with 2+ sessions on different days within 14 days	Usage	30%+
Feedback usefulness	% rating feedback "helpful" on a 3-point scale, with reason	Perception	70%+
Feedback accuracy flags	% of feedback items flagged "not accurate"	Quality (lower is better)	under 10%
Rubric movement	Share of retry pairs where a blind reviewer rates attempt 2 better on the target dimension	Learning signal	Report, no target yet

Targets are provisional placeholders for a first cohort and should be reset after the research phase. Session counts show engagement, not learning.

7. MVP scope and feature prioritisation

Rule applied: a feature ships in the P0 MVP only if the core loop breaks without it, or if privacy and safety require it.

Evaluation of the listed candidate features

Feature	Priority	Decision and rationale
Account creation and sign-in	P0	Needed to save history and delete data. Email magic link plus Google sign-in; no passwords to manage.
Goal-based onboarding	P0 (lite)	Two screens: goal and comfort mode. Drives scenario recommendations.
Scenario selection	P0	Library of 6 scenarios across 3 categories.
Voice responses	P0	Differentiating; needs a speech-to-text provider. Cut to P1 if research shows text-first preference.
Text responses	P0	Required for accessibility and privacy comfort.
Audio playback and recording controls	P0 (lite)	Record, pause, re-record, play back in-session. Playback of the local audio, not stored.
AI feedback	P0	The core value. Structured output, 2 strengths and 1–2 improvements.
Response retry and revision	P0	The loop's most important part.
Practice history	P0 (lite)	List of sessions with attempts. Transcripts only, no audio.
Basic progress tracking	P0 (lite)	Per-scenario attempt comparison plus qualitative notes. No overall score.
User settings and privacy controls	P0	Delete transcripts, delete account, retention choice. Cannot be deferred for a minors audience.
Responsive layouts	P0	Mobile-first; most users likely on phones.
Basic analytics	P0 (lite)	Event counts, no session replay, no audio content, no third-party ad trackers.

Prioritised feature list

Tier	Features
P0 (MVP)	Auth; lite onboarding; scenario library (6); scenario detail; practice room (text and voice); transcript review; AI feedback; revise and retry; session history; attempt comparison; settings, privacy and deletion; consent and age gate; empty, loading, error, permission-denied and offline states; basic analytics; abuse and cost controls
P1	Reminders and goals; more scenarios and difficulty levels; user-written custom scenarios; saved snippets ("phrases I like"); feedback language options; shareable summary for a mentor (user-controlled); facilitator cohort links; audio retention opt-in; PWA install
P2	Multilingual practice; live conversational role-play; pace and filler-word feedback (with careful fairness review); coach or mentor review integration; native apps; org dashboards

Prototype vs. functioning backend

Capability	Clickable Figma or HTML prototype	Functional MVP
Auth	Mock screens	Real auth provider
Scenario content	Hard-coded 3 scenarios	Database or content files
Voice recording	Simulated waveform animation	Real MediaRecorder capture
Transcription	Pre-written transcript	Speech-to-text API
AI feedback	Scripted feedback for 1–2 sample responses	Live LLM call with validated JSON
History and progress	Static sample data	Persisted data

Capability	Clickable Figma or HTML prototype	Functional MVP
Privacy controls	Screens only	Working deletion and retention jobs
Analytics	None	Event tracking

Recommendation: Run the research prototype with scripted feedback, but also test 3–5 real responses through a Wizard-of-Oz or live model call to check whether feedback quality, not just the flow, holds up.

8. Information architecture and sitemap

Public (signed out)

- Landing page
  - How it works, Scenarios preview, Privacy in plain language, FAQ
- Sign up / Sign in
- Legal: Privacy notice, Terms, Age and guardian information

App (signed in)

- Onboarding (first run only): welcome and age → consent → goal → comfort mode → first scenario suggestion
- **Home** (dashboard): suggested scenario, continue, recent attempts
- **Practise**
  - Scenario library (filter by category and goal)
  - Scenario details
  - Practice room (prompt, text or voice, transcript review)
  - Feedback
  - Revise and retry (compare attempts)
- **Progress**
  - History list; scenario detail with attempts
  - Attempt comparison
- **Me**
  - Profile, Preferences, Privacy and data, Accessibility, Delete account

Navigation model

- **Mobile:** bottom tab bar with three tabs: Home, Practise, Progress. The "Me" menu sits in a top-right avatar. During a practice session, the tab bar is hidden and replaced by a single "Leave practice" control to reduce distraction.
- **Desktop:** left rail with the same destinations plus Me; the practice room uses a centred single column, no sidebar.
- **Principles:** one primary action per screen; always a visible exit; no deep nesting beyond three levels; the user always knows whether they are speaking, typing or reviewing.

Entry points to the practice room: Home suggestion, Scenario details, "Try again" from Feedback, "Practise again" from History.

9. End-to-end user journeys

Journey 1: New user, onboarding to first practice

1. User arrives on the landing page and reads "How it works" (3 steps) and a plain-language privacy summary.
2. Taps **Start practising**; chooses email magic link or Google.
3. Age step: enters birth year. Under 13 is blocked. Ages 13–15 are blocked for MVP **[Assumption: launching 16+; confirm with legal review]**. Ages 16–17 see the guardian note and a shortened consent summary; 18+ see the standard consent.
4. Consent screen: what is processed (text, voice transcribed), who processes it (AI and speech providers), what is stored (transcripts only), and how to delete. Separate toggle: "Use my anonymised feedback ratings to improve SpeakHER" (off by default).
5. Goal: "What are you preparing for?" (multi-select: explaining my project, interviews, asking questions, networking, team discussions, not sure yet).
6. Comfort mode: "How would you like to practise?" Voice, text, or either. Can change later.
7. Recommended first scenario: "Explain your project in 60 seconds" with a preview.
8. Scenario details: audience, goal, time guide, tips. Taps **Start**.
9. Practice room: reads prompt; chooses voice or text; responds.
10. Voice only: transcript appears for review and correction; taps **Get feedback**.
11. Feedback screen appears with strengths and one or two improvements.
12. User taps **Try it again** or **Done for now**.
13. First-session survey (one tap): "Was this feedback helpful?"

Journey 2: Returning user repeats a scenario

1. Opens Home; sees "Continue" card for the last scenario and a suggestion.
2. Taps the history item or Scenario library; opens the scenario.
3. Details show "Your earlier attempts" and the focus point from last time.
4. Starts practice; the previous focus is shown as an optional reminder.
5. Submits a response; feedback shows what changed compared with the last attempt (specific, not a score).
6. Can compare attempts side by side in Progress.

Journey 3: Declines microphone access, chooses text

1. In the practice room, taps **Voice**; browser prompts for mic access. User denies.
2. A calm message appears: "No problem. You can type your answer instead. If you want to use voice later, you can enable the microphone in your browser settings." Shows a short "how to enable" link.
3. The text composer opens with the same prompt, timer and character guide; the choice is remembered as a preference (editable in Settings).
4. User submits typed response; feedback differs as defined in section 12 (text-specific dimensions).
5. No repeated mic prompts; voice remains available, not pushed.

Journey 4: Reviews feedback and submits an improved response

1. On Feedback: reads "What worked" (2 items with quoted evidence), "One thing to try" (1–2 items), and an optional "Try another way" hint.
2. Taps an item to see the evidence from her response.
3. Disagrees with one item: taps **This doesn't fit** and picks a reason (not accurate, not helpful, not relevant to my goal, other). The item is hidden and the signal is stored.
4. Taps **Revise my answer**: the original response is pre-filled (text) or the transcript is editable and the voice re-record is offered.
5. Edits or re-records, submits.
6. New feedback compares attempts: "Your explanation now names who the project helps."
7. Views the two attempts side by side; chooses to keep both, or delete one.

Journey 5: Manages or deletes practice data

1. Me → **Privacy and data**.
2. Sees plain-language summary: what is stored, retention setting, and which providers process responses.
3. Options: delete a single session; delete all transcripts and history; change retention (30 days / 1 year / until I delete); download my data; delete account.
4. Delete account: warning screen lists what will be erased and confirms; type "delete" to confirm; session ends.
5. Confirmation screen; email confirmation sent; deletion completed within the stated window (target 30 days, immediate for visible data) **[Validate with legal]**.

10. Screen-by-screen requirements

Every screen inherits these global rules: keyboard-operable, visible focus, 44×44 px minimum touch targets, support for 200% zoom and reduced motion, plain-language errors with a recovery action, and no content that depends on colour alone.

S1. Landing page

Aspect	Specification
Purpose	Explain what SpeakHER does and earn trust to sign up.
User goal	Understand in 15 seconds whether this is for me and is it safe.
Key elements	Hero with tagline "Find the words to express your thoughts."; a looping, reduced-motion-safe demo of a response getting a highlighted suggestion; 3-step "How it works"; scenario preview cards; privacy summary; FAQ; primary CTA "Start practising".
Actions	Sign up, Sign in, view privacy, expand FAQ.
Info displayed	Honest limits: "SpeakHER gives practice and feedback. It does not replace a mentor or guarantee outcomes."
States	Default; loading; offline notice.

Aspect	Specification
Errors	Failed asset load falls back to static illustration.
Mobile	Single column; sticky CTA; demo is a static card unless motion allowed.
Accessibility	One H1; skip link; demo has text alternative; FAQ as accessible disclosure widgets.

S2. Sign-up and authentication

Aspect	Specification
Purpose	Create or access an account with minimal data.
Elements	Email field with magic link; Continue with Google; link to Privacy and Terms.
States	Link sent; link expired; invalid email; rate-limited; network error.
Errors	"We couldn't send that link. Check the address and try again." Never reveal whether an email exists.
Mobile	Email keyboard type; auto-open mail app link.
Accessibility	Labels, error summary linked to field, focus returns to error.

S3. Onboarding and goal selection

Aspect	Specification
Purpose	Capture age, consent, goals and comfort mode in under 90 seconds.
Steps	Age (birth year) → Consent (privacy summary, separate toggle for research use) → Goals (multi-select chips) → Comfort mode → Suggested first scenario.

Aspect	Specification
Actions	Back, Skip goals, Continue. Skipping is allowed for goals only.
Info displayed	Plain-language privacy; progress indicator "Step 2 of 5".
Errors	Under-age: kind explanation and link to resources, no data stored.
Mobile	One question per screen; large chips.
Accessibility	Fieldsets and legends; progress announced; consent not pre-ticked.

S4. Dashboard (Home)

Aspect	Specification
Purpose	Get back into practice in one tap.
Elements	Greeting by first name; "Suggested for you" scenario card; "Pick up where you left off" card; last 3 attempts; small weekly practice summary (no streak).
Empty state	"Nothing here yet. Try your first scenario." with an illustrated notebook page.
Loading	Skeleton cards.
Errors	Inline retry; cached content if available.
Mobile	Cards stack; bottom tab bar.
Accessibility	Headings per section; cards are single links with clear names.

S5. Scenario library

Aspect	Specification
Purpose	Browse and choose a scenario.
Elements	Category tabs (Explain, Ask and Join, Interview and Intro); scenario cards with title, audience, time guide, difficulty; search (P1); filter by goal.
Empty state	"No scenarios match. Clear filters."
Mobile	Filter as a bottom sheet.
Accessibility	Filters announce result count.

S6. Scenario details and instructions

Aspect	Specification
Purpose	Set context before practice.
Elements	Title; situation ("You're at a community meetup. Someone asks what you're building."); audience; what to aim for (1–3 bullets); suggested length; optional tips; "Your earlier attempts"; mode choice; **Start** button.
Info	Time guide 60–90 seconds speaking or 100–150 words; no timer pressure.
Mobile	Sticky Start button.
Accessibility	Tips in a collapsible region; scenario text readable by screen reader.

S7. Practice room (voice and text)

Aspect	Specification
Purpose	A calm space to respond.

Aspect	Specification
Elements	Prompt card pinned at top (always visible); mode switch (Voice / Text) with equal prominence; text area with word count and soft guide; record control; optional notes area for bullet points; "Leave practice" control; gentle reminder "You can take your time."
Voice states	Idle; requesting permission; ready; recording (waveform + elapsed time); paused; recorded (playback); transcribing; transcript review; error.
Text states	Empty; typing; autosaved draft (local); too short (soft nudge); too long (soft limit).
Errors	Mic not found; permission denied; recording failed; transcription failed (offer retry or type instead); network lost (keep local draft).
Mobile	Large record button in thumb reach; prevent screen sleep while recording; handle app backgrounding by pausing and warning.
Accessibility	Record control is a button with state announced ("Recording, 00:12"); visible timer plus text status; captions for any audio; keyboard shortcut optional and documented.

S8. Recording, playback and submission

Aspect	Specification
Purpose	Let the user check their recording and transcript before it is evaluated.
Elements	Playback bar with play, scrub, speed; transcript editor with highlighted low-confidence words; "Re-record"; "Edit transcript"; **Get feedback**; note "Your audio is not saved after this step." (only if true in implementation).

Aspect	Specification
Behaviour	The AI evaluates the reviewed transcript, never the raw audio **[Recommendation]**.
Errors	Empty or silent recording detected; transcript confidence low prompts review.
Accessibility	Transcript is the accessible equivalent; editable with a keyboard.

S9. AI feedback

Aspect	Specification
Purpose	Deliver specific, digestible, respectful feedback.
Layout	(1) Short summary line; (2) "What worked" 1–3 cards with quoted evidence highlighted; (3) "One thing to try" 1–2 cards with an example of an alternative approach; (4) "Try another way" optional prompt; (5) actions.
Actions	Revise my answer; Try a new scenario; "This doesn't fit" per item; "Helpful / Not helpful"; view my full response.
States	Generating (stages: "Reading your answer…"); success; partial (some dimensions unavailable); failed (retry, keep response); content-flagged (supportive message).
Mobile	Cards stack; collapsed sections; sticky "Revise" button.
Accessibility	Highlights also marked by underline and text, not colour alone; headings per section; live region for completion.

S10. Revision and retry

Aspect	Specification
Purpose	Make improving the response the default next step.
Elements	Previous response (collapsible) beside a fresh editor on desktop, stacked on mobile; the improvement focus pinned ("Focus: say who it helps"); optional "Show an example approach" (collapsed by default).
Behaviour	The user decides whether to start fresh, edit the original, or re-record. Examples show structure, not wording to copy.
After submit	Feedback includes a "What changed" section.

S11. Practice history and progress

Aspect	Specification
Purpose	See attempts and how responses evolved.
Elements	Timeline of sessions; per-scenario page with attempts; attempt comparison view (two columns, highlighted additions); "Focus areas you've worked on" (derived from feedback tags); no total score, no streak.
Empty state	"Your practice will show up here."
Mobile	Compare as swipeable tabs.
Accessibility	Comparison available in a linear reading order.

S12. Profile, settings, privacy and account deletion

Aspect	Specification
Sections	Profile (name, avatar from preset set); Preferences (default mode, feedback depth); Accessibility (reduce motion, larger text, high contrast); Privacy and data (retention, delete sessions, download data, provider list, withdrawal of research consent); Account (sign out, delete account).
Deletion	Type-to-confirm; clear summary of what is erased; confirmation email.
Errors	Failed deletion shows a persistent banner with a retry and support contact.

S13. Empty, loading, error, permission-denied and offline states

State	Behaviour
Empty	Notebook-style illustration plus one next action.
Loading	Skeleton shapes; feedback generation uses staged messages and never a fake progress bar.
Error	What happened, what is saved, what to do. Example: "That didn't go through. Your answer is saved on this device. Try again."
Permission denied (mic)	Text mode offered; how-to-enable link; no repeated prompts.
Offline	Banner; local draft retained; browsing scenarios allowed if cached; submission disabled with explanation.

11. Functional requirements and acceptance criteria

IDs: **FR-** = functional requirement. All acceptance criteria are Given/When/Then and testable. Edge cases are listed per feature.

FR-1 Account creation and sign-in (P0)

- **Description / story:** As a user, I can create an account and sign in so my history is saved. Depends on: auth provider, email service.
- **Requirements:** email magic link and Google sign-in; session persists 30 days; sign-out on all devices; no password storage.
- **Behaviour:** after first sign-in, a user without consent records is routed to onboarding.

ID	Given	When	Then
FR-1.1	A valid email	The user requests a magic link	A single-use link valid for 15 minutes is sent and the UI confirms without revealing account existence
FR-1.2	An expired or used link	The user opens it	They see a friendly error and can request a new link
FR-1.3	5 link requests in 10 minutes from one email or IP	A 6th is attempted	The request is blocked and the user is told to wait
FR-1.4	A signed-in user	They sign out	The session is invalidated server-side and protected routes redirect to sign-in

Edge cases: email client blocks link; link opened on a different device; Google account without a verified email.

FR-2 Age gate, consent and goal onboarding (P0)

- **Description / story:** As a new user I give an age and consent in plain language and pick goals so I get relevant scenarios.
- **Requirements:** birth year captured, stored only as the age band (16–17 / 18+) after verification; consent versions recorded with timestamps; research-use toggle off by default; goals optional.

ID	Given	When	Then
FR-2.1	A user under 16	They submit the age step	The account is not created and no personal data is retained
FR-2.2	A user aged 16–17	They reach consent	The guardian note is shown and the privacy summary uses the age-appropriate version **[Validate legally]**
FR-2.3	A user on the consent step	They continue without ticking the required acknowledgment	They cannot proceed and the missing item is identified
FR-2.4	A completed onboarding	The user reaches the dashboard	Their goals and comfort mode appear in preferences and can be edited

Edge cases: user closes app mid-onboarding (resume at the last step); user changes age after sign-up (requires support review).

FR-3 Scenario library and details (P0)

- **Description / story:** As a user I choose a scenario and understand what is expected before I start.
- **Requirements:** 6 scenarios at launch, each with title, situation, audience, objective, tips, time guide, category, difficulty (1–2), and scoring focus tags; content stored as versioned data.

ID	Given	When	Then
FR-3.1	The library loads	The user filters by a category	Only matching scenarios display and the count is announced
FR-3.2	A scenario with earlier attempts	The user opens its details	Earlier attempts and the last focus point are shown
FR-3.3	No scenarios match filters	The list is empty	An empty state with "Clear filters" appears

Edge cases: scenario updated while a user is mid-session (session keeps the version it started with).

FR-4 Text response (P0)

- **Requirements:** text area with autosave to local storage every 5 seconds; soft minimum 15 words; soft maximum 400 words; paste allowed.

ID	Given	When	Then
FR-4.1	A user types a response	They refresh the page	The draft is restored from local storage
FR-4.2	A response under 15 words	The user submits	A gentle prompt suggests adding detail and offers "Submit anyway"
FR-4.3	A response over 400 words	The user submits	They are asked to shorten it, and the limit is explained

FR-5 Voice response, playback and transcript review (P0)

- **Requirements:** browser audio capture (MediaRecorder) with permission prompt triggered only by user action; max 3 minutes; pause and resume; playback before submission; audio processed in memory and sent for transcription over TLS; audio not retained after transcription **[Recommendation]**; transcript editable before feedback; low-confidence words highlighted.

ID	Given	When	Then
FR-5.1	Mic permission is not yet granted	The user taps Record	The browser prompt appears once with a short explanation beforehand
FR-5.2	Mic permission denied	The user returns to the practice room	Text mode is offered, and voice shows a "how to enable" help link without re-prompting
FR-5.3	A recording in progress	The 3-minute limit is reached	Recording stops, the audio is kept for review, and the user is told why

ID	Given	When	Then
FR-5.4	A completed recording	The user taps Get feedback	The AI evaluates the transcript as currently edited, not the earlier version
FR-5.5	A silent recording	Transcription returns empty	The user is told "We couldn't hear anything" and offered re-record or type
FR-5.6	Transcription fails	The service errors	The audio stays in memory for retry, up to 2 retries, then offers text entry
FR-5.7	The user leaves the room	They confirm leaving	In-memory audio is discarded

Edge cases: device with no mic; Bluetooth microphone changes mid-recording; tab backgrounded on iOS; very quiet speech; background noise; unsupported browser.

FR-6 AI feedback (P0)

- **Requirements:** feedback produced from the reviewed response, scenario context and user goal; output validated against a JSON schema; maximum one retry on invalid output; response in under 15 seconds at the 90th percentile **[Recommendation: tune after testing]**.

ID	Given	When	Then
FR-6.1	A valid response	Feedback is requested	1–3 strengths and 1–2 improvements are returned, each quoting or pointing to the user's own words
FR-6.2	A feedback item quotes text	It is rendered	The quote exists verbatim in the submitted response (validator check); otherwise the item is dropped
FR-6.3	A response under 15 words or off-topic	Feedback is requested	The system asks the user to add detail or states that it could not evaluate against the prompt, with no scores

ID	Given	When	Then
FR-6.4	The model times out	20 seconds elapse	A retry option is shown, the response is preserved, no partial feedback shown
FR-6.5	A user flags an item "This doesn't fit"	The flag is submitted	The item is hidden for that attempt and the flag is stored with a reason
FR-6.6	A response contains distress or safety signals	Feedback is requested	The normal feedback is replaced by a supportive message and resources (see section 14)

FR-7 Revision and retry (P0)

ID	Given	When	Then
FR-7.1	A feedback screen	The user taps Revise	The previous response is available and the focus point is pinned
FR-7.2	A second attempt submitted	Feedback is generated	It includes a "What changed" section comparing attempts and does not claim improvement unless evidenced by specific differences
FR-7.3	5 attempts on one scenario in one session	A 6th is requested	The user is asked if they want to take a break or try another scenario, with no hard block unless cost limits are hit

FR-8 Practice history and progress (P0)

ID	Given	When	Then
FR-8.1	A user with sessions	They open Progress	Sessions are listed newest first with scenario, date and attempt count

ID	Given	When	Then
FR-8.2	Two attempts on one scenario	The user chooses Compare	Both are shown side by side with highlighted differences
FR-8.3	A user with no sessions	They open Progress	An empty state links to the library
FR-8.4	A retention period expires	The cleanup job runs	Expired transcripts are deleted and users are not shown them

FR-9 Settings, privacy controls and deletion (P0)

ID	Given	When	Then
FR-9.1	A signed-in user	They delete a session	The session, responses and feedback are removed from the database immediately
FR-9.2	A user requests account deletion	They confirm by typing "delete"	The account and associated data are removed, and a confirmation email is sent
FR-9.3	A user changes retention to 30 days	They save	Data older than 30 days is scheduled for deletion within 24 hours
FR-9.4	A user requests their data	They tap Download	A JSON export of their account, sessions and responses is generated
FR-9.5	A user withdraws research consent	They toggle it off	Future ratings are excluded, and the change is logged

FR-10 Responsive layout and accessibility (P0)

ID	Given	When	Then
FR-10.1	A viewport of 360–1440 px	Any core screen renders	No horizontal scroll, and all actions are reachable

ID	Given	When	Then
FR-10.2	A keyboard-only user	They complete the core loop	Every control is reachable and operable with logical focus order
FR-10.3	A screen reader user	They record and submit	State changes (recording, transcribing, feedback ready) are announced
FR-10.4	Reduced motion is set in the OS	Any animated element renders	Motion is replaced by a static or minimal alternative

FR-11 Basic analytics (P0)

ID	Given	When	Then
FR-11.1	A user performs a tracked event	The event fires	Only event name, anonymous ID, timestamp, scenario ID and mode are recorded, with no response text or audio
FR-11.2	A user opts out	They toggle analytics off	No further non-essential events are recorded

P1 and P2 features (reminders, custom scenarios, mentor-share summary, cohort links, multilingual practice, and others) will receive the same story, requirement and Given/When/Then treatment when promoted into scope.